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DIVP Assignment No :- 3

Q1 Consider the images shown. The image on the right was obtained by.
a) multiplying the image on the left by $(-1)^{x+y}$
b) Computing the DFT
c) Taking the complex conjugate of the transform
d) Computing the inverse DFT
e) multiplying the real part of the result by $(-1)^{x+y}$. Explain (mathematically) why the image on the right appears as it does

→ Let the Original image be $f(x, y)$ of size $M \times N$
Let its 2-DFT be $F_f(u, v)$
where,

$$F_f(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

and inverse DFT is

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v)\, e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Assume that, M & N are even

a) Multiplication by $(-1)^{x+y}$

$$g(x,y) = f(x,y)\,(-1)^{x+y}$$

$$(-1)^{x} = e^{j\pi x}$$

$$(-1)^{y} = e^{j\pi y}$$

$$(-1)^{x+y} = e^{j\pi(x+y)}$$

$$g(x,y) = f F(x,y)\, e^{j\pi x} e^{j\pi y}$$

Using the Frequency shifting property

$$G(u,v) = F\left(u - \frac{M}{2},\ v - \frac{N}{2}\right)$$

Thus, multiplying the image by $(-1)^{x+y}$ shifts its Fourier Spectrum by

$$\left(\frac{M}{2}, \frac{N}{2}\right)$$

b) Computing the DFT

Now compute the DFT of $g(x,y)$

$$G(u,v) = F[g(x,y)]$$

from the above,

$$G(u,v) = F\left(u - \frac{M}{2},\ v - \frac{N}{2}\right)$$

The Spectrum has been shifted by

$$\left(\frac{M}{2}, \frac{N}{2}\right)$$

c) Taking the complex conjugate

$$H(u,v) = G(u,v)$$

important DFT property is

$$F^{-1}\{F^*(u,v)\} = f^*(-x,-y)$$

where the coordinates are interpreted modulo M and N

Therefore, when we take the complex conjugate of the transform and then perform the inverse DFT the spatial image becomes reflected about both axes

Since the original image is a real image

$$f^*(x,y) = f(x,y)$$
$$f^*(-x,-y) = f(-x,-y)$$

So image undergoes a 180° rotation

d) Computing the Inverse DFT

we now calculate

$$h(x,y) = F^{-1}\{G^*(u,v)\}$$

Using the conjugate property

$$h(x,y) = g^*(-x,-y)$$

Because the image is real

$$g^*(-x,-y) = g(-x,-y)$$

Recall that

$$g(x,y) = (-1)^{x+y} f(x,y)$$

Therefore

$$h(x,y) = (-1)^{-x-y} F(-x,-y)$$

$$(-1)^{-x-y} = (-1)^{x+y}$$

We get,

$$h(x,y) = (-1)^{x+y} F(-x,-y)$$

e) Multiplying the Real Part by $(-1)^{x+y}$

$$r(x,y) = (-1)^{x+y} h(x,y)$$

Substitute the expression for $h(x,y)$

$$r(x,y) = (-1)^{x+y} [(-1)^{x+y} F(-x,-y)]$$

$$r(x,y) = (-1)^{2x+2y} F(-x,-y)$$

$$(-1)^{2x+2y} = 1$$

we obtain

$$r(x,y) = F(-x,-y)$$

* Properties of DFT are used

1. Frequency shifting / Centering Property

$$g(x,y) = F(x,y)(-1)^{x+y}$$

then DFT becomes

$$G(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

2. Complex Conjugation Property

$$F(x,y) \leftrightarrow F(u,v)$$

$$F^*(u,v) \leftrightarrow F^*(-x,-y)$$
$$F^{-1}\{F^*(u,v)\} = F^*(-x,-y)$$

3. Real - Image Property

$$F^*(x,y) = F(x,y)$$
$$F^*(-x,-y) = F(-x,-y)$$

4. Multiplication Property / frequency shifting Again

$$(-1)^{x+y}(-1)^{x+y} = (-1)^{2x+2y} = 1$$
$$F(x,y) \longrightarrow F(-x,-y)$$

which is 180° rotation

Q2 Consider a 3 x 3 Spatial Kernel that average the four closet neighbour of a point (x,y) but excludes the point itself from the average

a) Find the equivalent filter transfer function $H(u,v)$ in frequency domain.
b) Show that your result is a lowpass filter transfer function.

→ Four closet neighbour of pixel (x, y)

Top : $(x-1, y)$

Bottom : $(x+1, y)$

Left : $(x, y-1)$

Right : $(x, y+1)$

3×3 Kernel

$$h(x,y) = \frac{1}{4}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

a) for a spatial filter, the frequency respone is obtained by taking the fourier transfarm of kernel

The four non-zero term are located at $(-1, 0)$, $(1, 0)$, $(0, -1)$, $(0, 1)$

Using the 2D Fourier transfarm

$$H(u,v) = \sum_x \sum_y h(x,y)\, e^{-j2\pi\left(\frac{ux}{m} + \frac{vy}{N}\right)}$$

Substitute the four non-zero values

1. $(x, y) = (-1, 0)$

$$h(x,y) = h(-1,0) = \frac{1}{4} e^{-j2\pi\left(\frac{-u}{m}\right)}$$

$$h(-1,0) = \frac{1}{4} e^{j2\pi u/m}$$

2. $(x, y) = (1, 0)$

$$h(1,0) = \frac{1}{4} e^{-j2\pi u/m}$$

3. $(x, y) = (0, -1)$

$$h(0,-1) = \frac{1}{4} e^{j2\pi v/N}$$

4. $(x, y) = (0, 1)$

$$h(0,1) = \frac{1}{4} e^{-j2\pi v/N}$$

$$H(u,v) = \frac{1}{4}\left[e^{j2\pi u/m} + e^{-j2\pi u/m} + e^{j2\pi v/N} + e^{-j2\pi v/N} \right]$$

By Euler's Identity

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta$$

We got

$$H(u,v) = \frac{1}{4}\left[2\cos\left(\frac{2\pi u}{M}\right) + 2\cos\left(\frac{2\pi v}{N}\right) \right]$$

$$\therefore H(u,v) = \frac{1}{2}\left[\cos\left(\frac{2\pi u}{M}\right) + \cos\left(\frac{2\pi v}{N}\right) \right]$$

b) i) Response at zero frequency

$u = 0, \quad v = 0$

$$H(0,0) = \frac{1}{2}\left[\cos(0) + \cos(0) \right]$$

$\cos(0) = 1, \quad H(0,0) = \frac{1}{2}(1+1) = 1$

$\therefore H(0,0) = 1$

This means the DC Component and very low frequency components are passed strongly

2) Behaviour at low Frequencies

$$\cos\theta = 1 - \frac{\theta^2}{2}$$

$$\cos\left(\frac{2\pi u}{M}\right) \approx 1$$

$$\cos\left(\frac{2\pi v}{N}\right) \approx 1$$

$$H(u, v) \approx 1$$

near the origin

Therefore,

low frequency components are passed with high gain

3) Effect on Rapid Image variations

high Spatial frequencies correspond to rapid changes in intensity such as;

- Sharp edges
- Fine details
- alternating pixel pattern

The averaging operation reduces many such rapid variations because each pixel is replaced by an average of its neighbours Thus, the filter performs a smoothing

Operation Smoothing is associated with low-pass filtering.

Therefore,

Four-neighbour averaging ⇒ Smoothing ⇒ low pass frequency emphasis.